



NDAI OPERATING SYSTEM / PRODUCT SENTINEL

THRAWN V2 GUIDE

Command harness, board mechanics,
heartbeats, proofs, and outputs.

SWIFTUI MAC APP

CODEX 5.5 EXTRA HIGH

PRODUCT SENTINEL

THRAWN + ARCHIVIST

MICROSOFT CLARITY SIGNALS

Generated 2026-05-21 15:08 CT from live Thrawn workspace state.

Design language follows NDAI dark-first tokens: obsidian, carbon, chrome, automation green.

Operating Picture

01

System As It Stands

Thrawn V2 is a Swift-native command harness with a clean two-agent roster, a Product Sentinel objective, proof storage, Citadel synthesis, and a board-driven delegation contract.

AGENT

Thrawn

Command layer. Core identity is Codex 5.5 Extra High; fallback state is visible but does not replace the identity.

AGENT

Samwell Tarly

Arch Maester. Maintains product pages and the rolling 72-hour brief from immutable evidence.

MISSION

Product Sentinel

Active objective watches Vaultage, Hit Zero, Cyclops, and Sandpro OMP.

Active Objectives

ID	PLAYBOOK	INPUT	PHASE	TASKS
OBJ-1779377589	product-sentinel	all products	phase 1	1/2

Important Truth

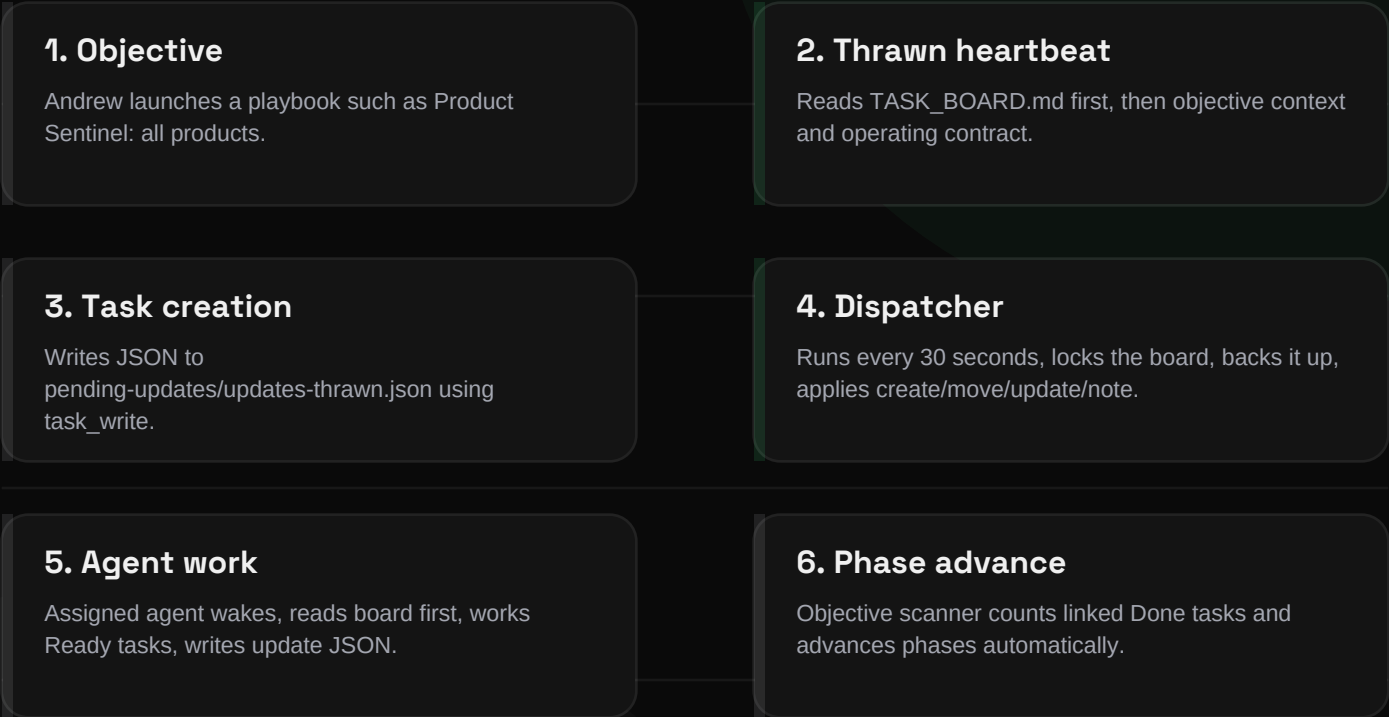
The architecture is board-driven. Agents receive the board first on every heartbeat, but the board itself is only mutated by the dispatcher. Thrawn and Samwell Tarly now have narrow task_write permission so they can submit JSON updates without directly editing the board.

Delegation & Board Mechanics

02

Delegation Flow

Objectives create intent. Thrawn decomposes intent into board tasks. Agents execute tasks. The dispatcher applies updates and the scanner advances objective phases.



Board Files

PATH	PURPOSE
TASK_BOARD.md	Human-readable Kanban source of truth
pending-updates/updates-*.json	Agent-submitted board mutations
TASK_BOARD.md.lock	Cross-process lock during dispatcher write
board-backups/	Pre-mutation board snapshots

Heartbeat Settings

03

Heartbeats

The scheduler is native Swift, not external cron. It checks every 30 seconds and persists last-run timestamps across app restarts.

AGENT	ID	CADENCE	HEARTBEAT	STATE
Thrawn	thrawn	every 15 min (:00/:15/:30/:45)	thrawn.HEARTBEAT.md	enabled
Samwell Tarly	archivist	:55 at 8 AM, 1 PM, 6 PM	archivist.HEARTBEAT.md	enabled

Briefings

SOD briefing fires at 7:00 AM. EOD briefing fires at 7:00 PM. BriefingService routes one-shot prompts through scheduler plumbing and logs generation in runtime events.

Reliability

Running agents are skipped until complete. Last-run timestamps prevent immediate duplicates after sleep, crash, or reopen. Watchdogs prevent hung heartbeat and dispatcher loops.

Product Sentinel Check Windows

WINDOW	TIME	TIMEZONE	DEDUPE
morning	08:10	America/Chicago	productId + yyyy-MM-dd + window
afternoon	13:10	America/Chicago	productId + yyyy-MM-dd + window
evening	18:10	America/Chicago	productId + yyyy-MM-dd + window

Agents, Tools, Routing

04

Agents Know The Board

Every heartbeat prompt includes TASK_BOARD.md first. The preamble tells the agent to find tasks where Owner equals its name and Status equals Ready.

AGENT	ROLE	TIER	TOOLS
Thrawn	Lead	premium	file_read, log_read, proof_read, wiki_read, task_write
Samwell Tarly	Memory & Proof	premium	file_read, log_read, proof_read, proof_write, wiki_read, wi...

Thrawn Core

Identity: Codex 5.5 Extra High. Routing tries OpenAI gpt-5.5 xhigh first. If unavailable, fallback is visible as degraded status without changing the core identity label.

Product Sentinel Route

The left selector controls background/Product Sentinel model route. It is not Thrawn's core brain. V2 experimental agents use explicit AgentSpec routing.

Permission Principle

Agents get the tools they need, not blanket shell access. task_write lets them submit board updates, while the dispatcher remains the only process that edits TASK_BOARD.md.

Product Sentinel Registry

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Products In Scope

The registry is the operating source for roots, commands, user flows, and Microsoft Clarity expectations.

PRODUCT	ID	ROOT	BUILD	TEST	CLARITY
Vaultage	vaultage	/Users/andrewemmel/Downloads/Vaultage	-	npm run test:e2e	expected
Hit Zero	hit-zero	/Users/andrewemmel/Desktop/apps/hitzero...	npm run build	npm run test	dashboard known
Cyclops	cyclops	/Users/andrewemmel/Desktop/apps/cyclops	npm run build	npm run lint	expected
Sandpro OMP	sandpro-omp	/Users/andrewemmel/Documents/New projec...	npm run build	npm run test:unit	expected

Microsoft Clarity Is A Major Signal

For products with clarity.expected = true, proof runs scan the repo for tracking code and write logs/microsoft-clarity.log. Recommendations should consult rage clicks, dead clicks, quick backs, excessive scrolling, recordings, heatmaps, funnels, smart events, and top affected pages.

Proofs, Citadel, Outputs

06

Output System

Thrawn separates immutable proof from readable context. Raw evidence stays in proof directories; Samwell Tarly keeps synthesis in Citadel pages.

ARTIFACT	LOCATION
Proof root	workspace/proofs/<product>/<yyyy-mm-dd>/<run-id>/
Metadata	proof-run.json
Verdict	verdict.md
Screenshots	screenshots/desktop-proof.png
Logs	logs/git-status.log, build.log, test.log, microsoft-clarity.log
Citadel product page	workspace/citadel/products/<product>.md
Rolling brief	workspace/citadel/rolling-72h.md
Runtime logs	workspace/logs/*.jsonl

Immutable Evidence

Raw proof directories should not be rewritten. New observations append to Citadel/context, preserving the original run.

Readable Context

The rolling 72-hour brief is intentionally short enough for Thrawn to read every cycle and use as context.

File System Map

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Live Paths

The app copies versioned OpsBundle files into Application Support, then runtime state evolves there.

THING	PATH
Active config	~/Library/Application Support/Thrawn/
Archive root	~/Library/Application Support/Thrawn Archives/
Workspace docs	workspace/*.md
Agents	workspace/agents/
Heartbeats	workspace/ops/heartbeats/
Board	workspace/ops/TASK_BOARD.md
Pending updates	workspace/ops/pending-updates/
Product registry	workspace/product-sentinel/products.json
Proofs	workspace/proofs/
Citadel	workspace/citadel/
Guide	workspace/THRAWN_GUIDE.md

Current Limitation / Next Build

Product Sentinel has registry, schedule, proof hook, Clarity scanning, and Samwell Tarly synthesis. The next natural step is a dedicated Sentinel runner/agent that autonomously starts products, drives browser flows, captures screenshots, and runs each scheduled window.

Runbook

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How To Operate It

Short, durable actions for the next build cycle.

Launch objective

Use Objectives > Product Sentinel with a product or all products.

Watch board

The board should fill after Thrawn heartbeat creates phase tasks.

Manual proof

Run `product-sentinel-proof.py --product <id>` for immediate evidence.

Review Clarity

Before UX changes, inspect Clarity dashboard signals and record project IDs.

Add agents slowly

Each new agent needs mandate, cadence, tool list, output paths, and review standard.

Promote Sentinel

Next durable role should own scheduled product checks and browser proof capture.